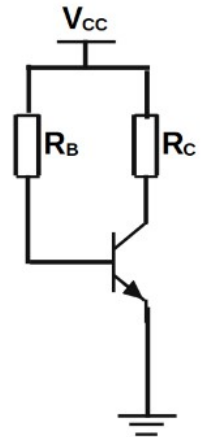


Tutorial series N°2

Problem 1 :

Consider the circuit opposite. Where: $V_{CC}=15\text{ V}$, $R_B=2\text{ M}\Omega$, $R_C=10\text{ K}\Omega$, $V_{BE}=0.7\text{ V}$, and $\beta=100$.

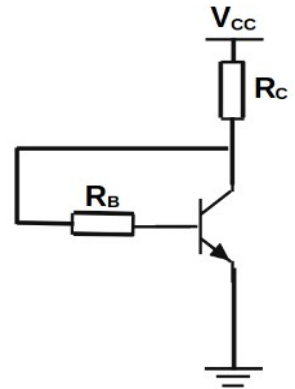
1. Calculate I_B , I_C , and V_{CE}
2. We add $R_E=1\text{ K}\Omega$ between the emitter and the ground, Calculate I_E , I_C , V_E , V_C , V_{CE} and V_B



Problem 2:

Consider the circuit opposite. Where: $V_{CC}=10\text{ V}$, $R_C=250\Omega$, $V_{BE}=0.6\text{ V}$, and $\beta=50$.

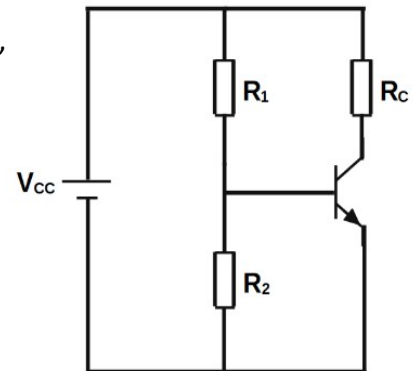
1. Determine the static load line.
2. We require a quiescent point in the midway of the preceding line; calculate R_B .



Problem 3:

Consider the circuit opposite. Where: $V_{CC}=12\text{ V}$, $R_1=160\text{ k}\Omega$, $R_2=16\text{ k}\Omega$, $R_C=1\text{ k}\Omega$, $V_{BE}=0.7\text{ V}$, and $\beta=100$.

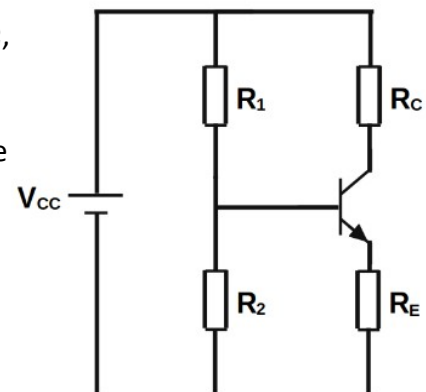
1. Determine the static drive line equation.
2. Determine the static load line equation.
3. Determine the quiescent point (**Q-point**) of the transistor.



Problem 4:

Consider the circuit opposite. Given: $V_{CC}=15\text{ V}$, $R_1=100\text{ k}\Omega$, $R_2=10\text{ k}\Omega$, $R_C=10\text{ k}\Omega$, $R_E=1\text{ k}\Omega$, $V_{BE}=0.7\text{ V}$, and $\beta=100$.

1. Perform the transformation of the biasing circuit as seen from the base of the transistor into its Thévenin equivalent.
2. Establish the expression I_E .
3. Calculate I_C , V_E , V_C , V_{CE} , and V_B .



Problem 5

Consider the circuit shown in the figure opposite with : $V_{CC}=11V$, $R_C=2.7k\Omega$, $R_E=390\Omega$, $\beta=135$, $h_{11}=2.7k\Omega$, and $h_{12}=h_{22}=0$.

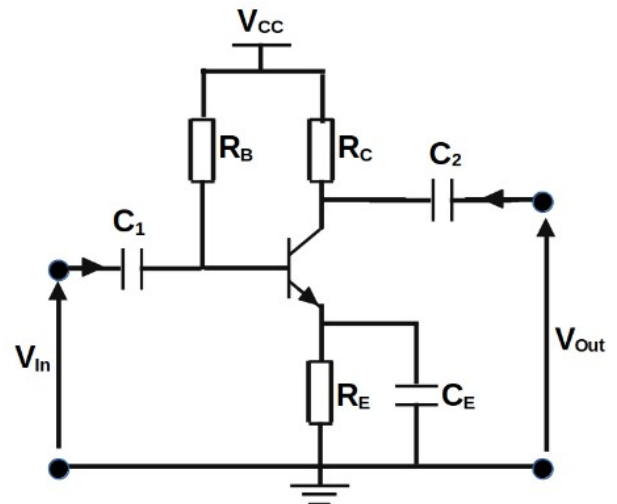
Static analysis:

Given $V_{BE}=0.7V$ and $V_C=5.6V$ (voltage between the collector and the ground).

1. Find the value of R_B .
2. Calculate the values of I_{C_0} and V_{CE_0}

Dynamic analysis:

3. Draw the equivalent circuit diagram.
4. Calculate the input resistance and the voltage gain.



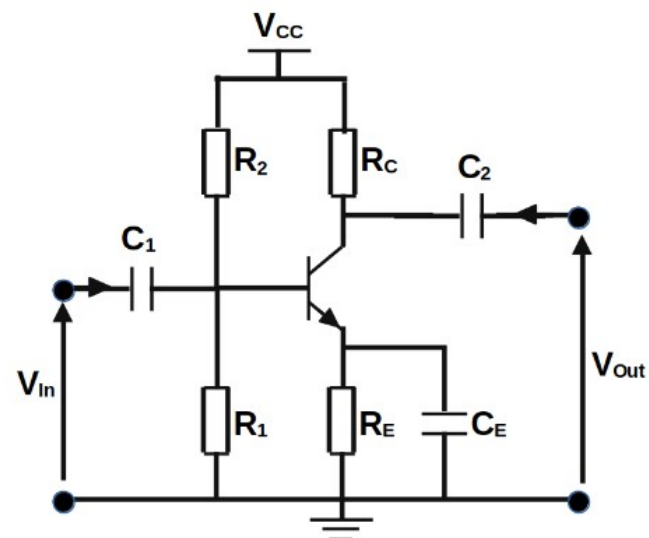
Problem 6:

Let's consider the amplifier as in the figure on the right. Where: $I_C=I_E=2.8mA$, $V_{BE}=0.6V$, $V_{CC}=24V$, $h_{11} \ll R_B (R_1//R_2)$, $h_{12}=h_{22}=0$, $h_{21}=\beta=100$.

1. Given that the amplifier has a no-load voltage gain of -220, an input impedance of 900Ω , and an operating point in the middle of the static load line, calculate the values of the resistors R_C and R_E
2. Find R_1 and R_2 if $R_B=10R_E$.

A load resistance (R_L) is placed between the collector and the ground at the output of the circuit.

3. Determine the gain in current and the output resistance.



Good Luck

“Together we learn, innovate, and succeed”